

1 Question 1: The Mandelbrot Set

In this question, we were asked to, given a range of x and y values in the complex plane, iterate over the quadratic equation: $z_{i+1} = z_i^2 + c$, where $c \in \{x + iy \mid x, y \in (-2, 2)\}$.

1.1 Methods

The required iteration was done through defining the function *complex_iterations*, which took the following parameters: *x_min*, *y_min*, *x_max*, *y_max*, (optional parameter) *test_pts*, and (optional parameter) *iterations*. Non-optional parameters are to define the bounds of the complex plane. The optional parameter, *test_pts*, is used to determine the number of sample points along each axis, while *iterations* specifies the maximum number of iterations performed.

With these parameters, we create our linspace and define a standard grid, where X is the real part of a complex number, c , and Y is the imaginary. To determine what points remain bounded, and which run off to infinity, we iterate for i in *iterations*. For each iteration, the recurrence relation

$$z_{i+1} = z_i^2 + c$$

is applied across the complex grid. After every update, points with magnitude greater than 2 are marked as diverged in a boolean array, and the iteration at which this occurs is stored, while points that never diverge are classified as bounded.

1.2 Findings

Using 500 test points and 10,000 iterations, the image produced by *complex_iterations* is the characteristic Mandelbrot fractal.

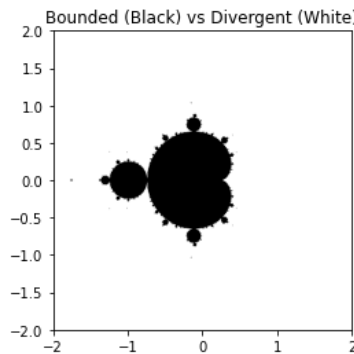


Figure 1: The Mandelbrot Set generated with 10,000 iterations.

2 Question 2: Lorenz Equations

In this question, we were asked to create a function that defines three Lorenz equations, solve said equations using SciPy's `solve_ivp`, and plot the result.

2.1 Methods

Lorenz' equations were defined in `lorenz_eqns`, a function dependent on time, state, and with three constant parameters: σ , b , and r . The following three equations were defined:

$$\dot{X} = -\sigma(X - Y) \quad (1)$$

$$\dot{Y} = rX - Y - XZ \quad (2)$$

$$\dot{Z} = -bZ + XY \quad (3)$$

These equations were then solved using `solve_ivp` from the SciPy library.

2.2 Findings

Plotting the y -component of the IVP solution, W , using three successive intervals of 1000 iterations and Lorenz' discrete time variable,

$$N = \frac{t}{\Delta t}, \quad \Delta t = 0.01,$$

successfully reproduced Figure 1 of Lorenz' *Deterministic Nonperiodic Flow*, shown in Figure 2.

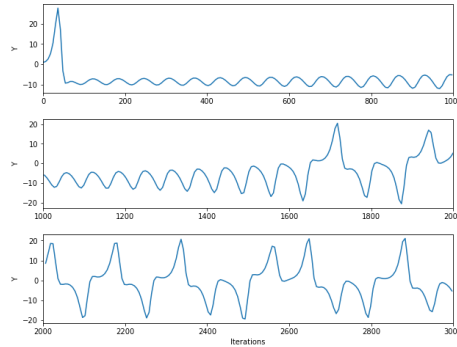


Figure 2: Reproduction of Lorenz' Figure 1

Projecting the solution onto the YZ and XY planes reproduced the characteristic structure shown in Figure 2 of Lorenz' *Deterministic Nonperiodic Flow*, as displayed in Figure 3.

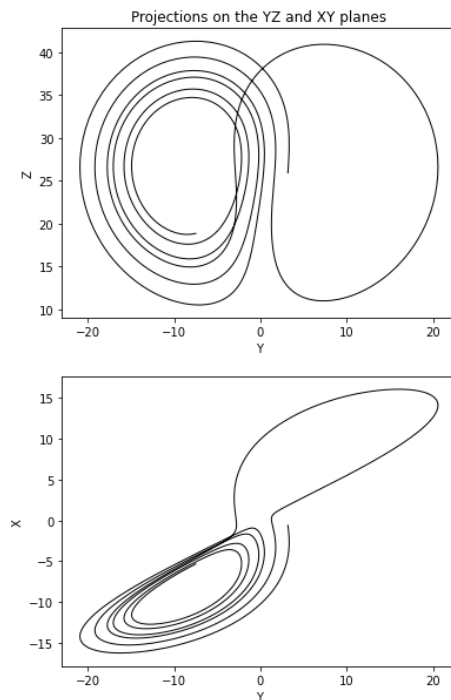


Figure 3: Reproduction of Lorenz' Figure 2

Using a slightly different initial condition produced a second solution, W' . The semi-logarithmic plot of the separation distance between W' and W is shown in Figure 4, illustrating the sensitive dependence on initial conditions.

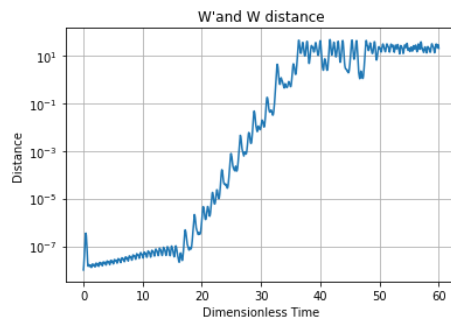


Figure 4: Semi-logarithmic plot of the separation between W' and W